

## Chapter 5

# TreeSwap: Efficient Swap Regret Minimization

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So far, we have seen that no-regret learning, and efficient equilibrium computation, are essentially possible whenever the set of deviations  $\Phi$  is “efficiently representable”, in particular, whenever *external* regret minimization is possible over  $\Phi$ . What about beyond? In particular, we have not yet addressed the *strongest* possible notion of  $\Phi$ -regret, namely, the case where  $\Phi$  contains *all* functions  $\varphi : \mathcal{X} \rightarrow \mathcal{X}$ —known as *swap regret*.

The fact that algorithms for  $\Phi$ -regret minimization grow increasingly complex as  $\Phi$  itself grows in complexity may lead one to believe that efficient swap regret minimization is hopeless. For example, the set of all functions  $\varphi : \mathcal{X} \rightarrow \mathcal{X}$  has *knfinite* dimension, so the techniques we have covered so far do not work. If  $\mathcal{X}$  is a polytope, we could run the Blum-Mansour algorithm over its vertices—this would indeed succeed, but it has the drawback that the number of vertices of  $\mathcal{X}$  is generally not polynomial in the dimension  $d$  of  $\mathcal{X}$ , and hence the algorithm is not efficient. Hence, for a long time, the possibility of swap regret minimization—and correspondingly, correlated equilibrium computation beyond normal-form games—remained a major open question.

The question has since essentially been fully resolved. In a simultaneous major breakthrough, Peng and Rubinstein [PR24] and Dagan, Daskalakis, Fishelson and Golowich [Dag+24], using essentially an identical algorithm now known as **TreeSwap**, showed that, if there is a regret minimizer on  $\mathcal{X}$  that achieves *external* regret  $\epsilon$  after  $M$  rounds, then there is a regret minimizer on  $\mathcal{X}$  that achieves  $\epsilon T$  *swap* regret after  $M^{1/\epsilon}$  rounds, and therefore efficient swap regret minimization *is* in fact possible, at least when  $\epsilon$  is a constant. They also showed nearly-matching lower bounds for normal-form games, which Daskalakis, Farina, Golowich, Sandholm and Zhang [Das+24] later extended to extensive-form games as well, thereby precluding the possibility of  $\text{poly}(d, 1/\epsilon)$ -time algorithms for swap regret beyond normal-form games. In this chapter, we will go over the upper bound.

## 5.1 The TreeSwap Algorithm

We now describe the **TreeSwap** algorithm, following Peng and Rubinstein [PR24] and Dagan, Daskalakis, Fishelson and Golowich [Dag+24]. Let  $\mathcal{R}$  be a no-regret algorithm on  $\mathcal{X}$  whose

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regret is bounded by  $\epsilon T$  after  $M$  rounds. Of course,  $\mathcal{R}$  will not necessarily have small swap regret. So, how can we hope to bound swap regret?

One thing to try might be the following. Instead of having  $\mathcal{R}$  update its strategy on *every* iteration, it will be lazy, and only update its strategy every  $M$  iterations, using the average utility on those  $M$  iterations. That is, at time  $M$ , it receives utility  $\frac{1}{M} \sum_{t=1}^M \mathbf{u}^{(t)}$ , at time  $2M$  it receives utility  $\frac{1}{M} \sum_{t=M+1}^{2M} \mathbf{u}^{(t)}$ , and so on. Then, between those  $M$  iterations, that is, while  $\mathcal{R}$  is playing the same strategy, we initialize *another copy of the same regret minimizer*, which we will call  $\mathcal{R}'$ , which updates its strategy on *every* iteration.

The purpose of  $\mathcal{R}'$  is to bound the value of the most profitable deviation from every single strategy that  $\mathcal{R}$  plays. Therefore,  $\mathcal{R}'$  can ensure that no strategy played by  $\mathcal{R}$  incurs large swap regret. Of course,  $\mathcal{R}'$  itself will have high swap regret... but this can be solved by introducing yet more layers!

The **TreeSwap** algorithm uses  $K$  layers of regret minimizers  $\mathcal{R}_0, \dots, \mathcal{R}_{K-1}$ , each responsible for ensuring that there is no profitable swap deviation for the regret minimizers in the next layer.  $\mathcal{R}_0$  updates on every iteration. Each regret minimizer  $\mathcal{R}_k$  updates  $M$  times slower than the previous regret minimizer  $\mathcal{R}_{k-1}$ , and resets every  $M$  updates. Hence, this algorithm runs for  $M^K$  steps. The strategy output by **TreeSwap** is the uniform distribution (not to be confused with the average) over the strategies  $(\mathbf{x}_0^{(t)}, \dots, \mathbf{x}_{K-1}^{(t)})$  output by the  $K$  regret minimizers. For simplicity of notation, we will henceforth zero-index timesteps and use the notation  $[n] = \{0, \dots, n-1\}$ .

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**Algorithm:** TreeSwap

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**Input:** Regret minimizers  $\mathcal{R}_1, \dots, \mathcal{R}_K$

**for** timesteps  $t \in [T] := [M^K]$  **do**

**for** layers  $k \in [K]$  **do**

**if**  $M^{k+1} \mid t$  **then** reset  $\mathcal{R}_k$

**else if**  $M^k \mid t$  **then** pass to  $\mathcal{R}_k$  the average of the last  $M^k$  utilities,

$$\frac{1}{M^k} \sum_{\tau=t-M^k}^{t-1} \mathbf{u}^{(\tau)}$$

$\mathbf{x}_k^{(t)} \leftarrow$  current strategy of  $\mathcal{R}_k$

    play mixed strategy  $\text{Unif}(\mathbf{x}_0^{(t)}, \dots, \mathbf{x}_{K-1}^{(t)}) \in \Delta(\mathcal{X})$

    receive utility  $\mathbf{u}^{(t)} \in \mathbb{R}^d$

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**Theorem 5.1.** The regret of **TreeSwap** is bounded by  $T \cdot (\epsilon + 1/K)$ . In particular, taking  $K = 1/\epsilon$  and  $M = \text{poly}(d, 1/\epsilon)$  (as is achieved by any reasonable external-regret-minimizing algorithm), **TreeSwap** achieves swap regret  $\epsilon T$  after  $M^K = d^{\tilde{O}(1/\epsilon)}$  iterations.

**Remark 5.2.** Theorem 5.1 only applies when  $T$  is a power of  $M$ . Handling the case where  $T$  is not a power of  $M$  requires a bit of care and results in a looser bound of  $T \cdot (\epsilon + 3/K)$ , but the main ideas are captured by the above result and its proof. [Dag+24]

**Remark 5.3.** We are mainly interested in the implications of Theorem 5.1 for its implications for swap regret in high-dimensional settings. However, it is worth noting that Theorem 5.1 achieves a new result even in the normal-form setting: its regret bound for normal form, when instantiated with multiplicative weights, is  $\log(N)^{\tilde{O}(1/\epsilon)}$ , which, for large  $N$  and  $\epsilon$ , is better than the bound of Blum-Mansour.

We now sketch a proof of Theorem 5.1.

*Proof Sketch.* In this proof, timesteps are zero-indexed, and  $[n] := \{0, \dots, n-1\}$ . Let  $\varphi : \mathcal{X} \rightarrow \mathcal{X}$  be any function. Then the (time-averaged) regret against  $\varphi$  is given by

$$\begin{aligned} \frac{1}{T} \text{Reg}(T, \varphi) &= \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}_{t \in [T]} \langle \mathbf{u}^{(t)}, \varphi(\mathbf{x}_k^{(t)}) \rangle - \langle \mathbf{u}^{(t)}, \mathbf{x}_k^{(t)} \rangle \\ &= \frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E}_{t \in [T]} \langle \mathbf{u}^{(t)}, \varphi(\mathbf{x}_k^{(t)}) \rangle - \frac{1}{K} \sum_{k=1}^K \mathbb{E}_{t \in [T]} \langle \mathbf{u}^{(t)}, \mathbf{x}_{k-1}^{(t)} \rangle \\ &= \frac{1}{K} \sum_{k=1}^{K-1} \underbrace{\mathbb{E}_{t \in [T]} \langle \mathbf{u}^{(t)}, \varphi(\mathbf{x}_k^{(t)}) - \mathbf{x}_{k-1}^{(t)} \rangle}_{\leq \epsilon} + \frac{1}{K} \underbrace{\mathbb{E}_{t \in [T]} \langle \mathbf{u}^{(t)}, \varphi(\mathbf{x}_0^{(t)}) - \mathbf{x}_{K-1}^{(t)} \rangle}_{\leq 1} \\ &\leq \epsilon + \frac{1}{K} \end{aligned}$$

where, in the second-to-last line, the bound on the first term comes from the regret bound of each phase between resets (of length  $M$  updates), noting that  $\varphi(\mathbf{x}_k^{(t)})$  is constant within any given phase and thus the regret against  $\varphi(\mathbf{x}_k^{(t)})$  is bounded by the external regret during the phase.  $\square$

## Bibliography for this chapter

- [PR24] B. Peng and A. Rubinstein. (2024). Fast Swap Regret Minimization and Applications to Approximate Correlated Equilibria. *Symposium on Theory of Computing (STOC)*.
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- [Das+24] C. Daskalakis, G. Farina, N. Golowich, T. Sandholm and B. H. Zhang. (2024). A Lower Bound on Swap Regret in Extensive-Form Games. *arXiv preprint arXiv:2406.13116*.